

Carbon Dioxide, Total

Test Details

METHODOLOGY

Enzymatic

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Use

Evaluate the total carbonate buffering system in the body, acid-base balance. High results may represent respiratory acidosis with CO₂ retention, or metabolic alkalosis (eg, prolonged vomiting). Low value may indicate respiratory alkalosis as in hyperventilation or metabolic acidosis (eg, diabetes with ketoacidosis).

Special Instructions

Maintain specimen in a tightly-stoppered tube.

Limitations

Interpretation requires clinical information and the other electrolytes.

Custom Additional Information

“Total carbon dioxide” consists of CO_2 in solution or bound to proteins, HCO_3^- , CO_3^{2-} , and H_2CO_3 . In practice, 80% to 90% is present as bicarbonate (HCO_3^-). “Hypercapnia” means excessive carbon dioxide in the blood. Impaired elimination of CO_2 reflects interaction of abnormalities in respiratory drive, the muscles of respiration, and the function of the lung. Elimination of carbon dioxide from the lung involves alveolar ventilation but not dead-space ventilation. Partitioning of these spaces is expressed as a ratio between dead space and total volume per breath: the tidal volume. The tidal volume normally is <0.30 . These and other aspects of pulmonary gas exchange, ventilation and their consequences are addressed as the partial pressure of arterial carbon dioxide, PaCO_2 , a part of arterial blood gases.¹

Note: Total CO_2 (Bicarbonate) results should be interpreted with **caution**, because escape of dissolved CO_2 from the sample prior to analysis is inevitable and will vary among laboratory locations. This is because the measured total CO_2 in serum decreases relative to the amount of time that the sample is open to the atmosphere. CO_2 loss occurs the moment the stopper is removed from the vacutainer tube. Labcorp facilities follow rigorous quality processes in order to reduce the exposure of patient samples to the atmosphere and minimize CO_2 loss, but even minor logistical variations at our laboratory locations may cause differences in the degree of CO_2 loss.

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

Separate serum or plasma from cells within 45 minutes of collection.

Stability Requirements

Temperature	Period
Room temperature	14 days
Refrigerated	14 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Reference Range

See table.

Age	Range (mmol/L)
0 to 30 d	16–29
31 d to 1 y	15–25
2 to 5 y	17–26
6 to 12 y	19–27
>12 y	20–29

Storage Instructions

Room temperature

Causes for Rejection

Improper collection and storage; improper labeling

Footnotes

1. Weinberger SE, Schwartzstein RM, Weiss JW. Hypercapnia. *N Engl J Med*. 1989 Nov 2; 321(18):1223-1231. PubMed 2677729

(<http://www.ncbi.nlm.nih.gov/pubmed/2677729>)